SQL Mini Project by Neetu

Level 1: Basics

1. Retrieve customer names and emails for email marketing

This helps the marketing team extract basic customer contact details for campaigns.

```
Select name, email from customers;
```

name	email
Thomas Owens	user1@example.com
Charles Grant	user2@example.com
Kaitlin Richards	user3@example.com
Christina Williams	user4@example.com
David Allen	user5@example.com
Mark Duke	user6@example.com
Briana Wright	user7@example.com
John Bryan	user8@example.com
Jason Thompson	user9@example.com
Shawn Hill	user10@example.com
Walter Jenkins	user11@example.com
Mary Knight	user12@example.com
Leslie Wilson	user13@example.com
Deborah Arias	user14@example.com
Austin Flores	user15@example.com
Amy Landry	user16@example.com
Randy Mooney	user17@example.com
Jeffrey Bray	user18@example.com
Amanda Bright	user19@example.com
Megan Lee	user20@example.com
Jessica Zamora	user21@example.com
Peter Phillips	user22@example.com
Kathryn Mathews	user23@example.com
Brandy Wright	user24@example.com
Cindy Hart	user25@example.com
Victoria Hall	user26@example.com
Adrienne Green	user27@example.com
Joseph Stuart	user28@example.com
Kara Zavala	user29@example.com
Victoria Acevedo	user30@example.com

2. View complete product catalog with all available details
The product manager may want to review all product listings in one go.

```
select * from products;
```

product_id	name	category	price	stock_quantity	added_on
1	Plant No	Home	639.43	152	2024-01-30 06:30:53
2	Population Social	Clothing	4813.68	84	2025-05-30 10:02:50
3	Available Answer	Electronics	2529.51	101	2025-04-13 01:11:46
4	Any Question	Clothing	4759.28	179	2025-06-03 13:34:03
5	Natural Network	Toys	4722.66	75	2023-11-06 00:47:37
6	If Whatever	Electronics	177.40	64	2024-12-19 10:37:14
7	Response Indeed	Clothing	4897.36	36	2025-03-29 02:43:08
8	Every Amount	Home	4173.60	156	2025-04-30 03:11:10
9	Common Study	Toys	985.19	171	2023-07-20 13:06:42
10	Development System	Electronics	4801.78	153	2025-03-12 08:22:57
11	Build Her	Books	1852.64	150	2024-09-08 01:09:15
12	Action Ask	Electronics	4017.01	19	2025-02-14 03:38:06
13	Full West	Books	2112.33	172	2023-09-15 03:13:38
14	Listen Development	Home	296.17	132	2024-10-12 11:49:26
15	Place Low	Electronics	723.97	46	2023-07-05 14:36:07
16	Special Fact	Toys	1094.18	15	2025-03-17 10:42:40
17	Everything Plant	Books	2496.68	120	2023-10-08 20:11:55
18	Some Them	Toys	3673.86	110	2024-07-25 00:33:53
19	Build High	Clothing	4707.14	47	2023-09-01 02:50:01
20	Real Source	Books	4398.66	197	2025-02-08 10:28:27
21	Bed Including	Clothing	3845.31	92	2025-01-21 11:52:40
22	Move Small	Books	4168.08	22	2023-11-12 18:26:30
23	Life Series	Clothing	2208.99	136	2024-03-09 09:19:13
24	Assume Serve	Home	3437.81	10	2024-07-10 05:05:17
25	South Free	Clothing	3543.58	44	2023-07-25 04:13:43
26	Movement Future	Electronics	3502.62	79	2024-01-23 08:39:07
27	Answer But	Home	1016.68	90	2025-04-06 21:54:11
28	East Foot	Toys	2414.93	128	2025-05-11 16:02:29
29	Drop Remember	Electronics	4160.45	89	2023-12-15 12:31:11
30	Future Sort	Home	3043.67	100	2025-05-17 06:55:28
31	Series Page	Electronics	2070.37	83	2024-04-01 00:24:06
32	Factor Where	Clothing	2295.14	167	2023-07-07 00:59:39
33	Involve Officer	Clothing	4244.09	112	2024-03-22 11:36:17
34	Despite Win	Electronics	1340.34	64	2024-11-27 06:55:45
35	Fire Often	Electronics	4734.89	198	2023-10-22 04:15:51
36	Television Stock	Clothing	421.73	115	2024-04-23 23:54:15
37	Between Up	Toys	1429.45	61	2025-01-27 01:30:37
38	Study Total	Toys	4413.68	31	2023-11-08 08:25:07
39	Data Add	Books	4063.38	62	2025-05-26 23:39:16
40	Toward Minute	Clothing	857.85	136	2025-04-04 07:04:21

product_id	name	category	price	stock_quantity	added_on
41	Serious Recognize	Electronics	4523.10	112	2023-08-20 06:16:55
42	Weight Enter	Clothing	3875.18	60	2024-01-26 08:21:55
43	Least Green	Toys	1398.26	81	2024-04-10 15:23:33
44	Actually Term	Electronics	396.11	85	2023-11-02 13:09:20
45	Suffer We	Electronics	3657.43	113	2024-07-25 12:40:57
46	Age Treatment	Home	3728.83	160	2023-12-01 10:51:52
47	Southern Thing	Electronics	512.46	40	2024-02-28 17:57:38
48	Group But	Clothing	4180.23	141	2023-11-29 01:38:12
49	Rest Something	Books	3077.82	181	2023-08-09 03:42:42
50	Stock Article	Home	834.75	131	2024-11-27 15:58:32
NULL	NULL	NULL	NULL	NULL	NULL

3. List all unique product categories
   Useful for analysing the range of departments or for creating filters on the website.

```
select distinct category
from products;
```

category
Home
Clothing
Electronics
Toys
Books

4. Show all products priced above ₹1,000

This helps identify high-value items for premium promotions or pricing strategy reviews.

```
select * from products

where price > 1000;
```

product_id	name	category	price	stock_quantity	added_on
2	Population Social	Clothing	4813.68	84	2025-05-30 10:02:50
3	Available Answer	Electronics	2529.51	101	2025-04-13 01:11:46
4	Any Question	Clothing	4759.28	179	2025-06-03 13:34:03
5	Natural Network	Toys	4722.66	75	2023-11-06 00:47:37
7	Response Indeed	Clothing	4897.36	36	2025-03-29 02:43:08
8	Every Amount	Home	4173.60	156	2025-04-30 03:11:10
10	Development System	Electronics	4801.78	153	2025-03-12 08:22:57
11	Build Her	Books	1852.64	150	2024-09-08 01:09:15
12	Action Ask	Electronics	4017.01	19	2025-02-14 03:38:06
13	Full West	Books	2112.33	172	2023-09-15 03:13:38
16	Special Fact	Toys	1094.18	15	2025-03-17 10:42:40
17	Everything Plant	Books	2496.68	120	2023-10-08 20:11:55
18	Some Them	Toys	3673.86	110	2024-07-25 00:33:53
19	Build High	Clothing	4707.14	47	2023-09-01 02:50:01
20	Real Source	Books	4398.66	197	2025-02-08 10:28:27
21	Bed Including	Clothing	3845.31	92	2025-01-21 11:52:40
22	Move Small	Books	4168.08	22	2023-11-12 18:26:30
23	Life Series	Clothing	2208.99	136	2024-03-09 09:19:13
24	Assume Serve	Home	3437.81	10	2024-07-10 05:05:17
25	South Free	Clothing	3543.58	44	2023-07-25 04:13:43
26	Movement Future	Electronics	3502.62	79	2024-01-23 08:39:07
27	Answer But	Home	1016.68	90	2025-04-06 21:54:11
28	East Foot	Toys	2414.93	128	2025-05-11 16:02:29
29	Drop Remember	Electronics	4160.45	89	2023-12-15 12:31:11
30	Future Sort	Home	3043.67	100	2025-05-17 06:55:28
31	Series Page	Electronics	2070.37	83	2024-04-01 00:24:06
32	Factor Where	Clothing	2295.14	167	2023-07-07 00:59:39
33	Involve Officer	Clothing	4244.09	112	2024-03-22 11:36:17
34	Despite Win	Electronics	1340.34	64	2024-11-27 06:55:45
35	Fire Often	Electronics	4734.89	198	2023-10-22 04:15:51
37	Between Up	Toys	1429.45	61	2025-01-27 01:30:37
38	Study Total	Toys	4413.68	31	2023-11-08 08:25:07
39	Data Add	Books	4063.38	62	2025-05-26 23:39:16
41	Serious Recognize	Electronics	4523.10	112	2023-08-20 06:16:55
42	Weight Enter	Clothing	3875.18	60	2024-01-26 08:21:55
43	Least Green	Toys	1398.26	81	2024-04-10 15:23:33
45	Suffer We	Electronics	3657.43	113	2024-07-25 12:40:57
46	Age Treatment	Home	3728.83	160	2023-12-01 10:51:52
48	Group But	Clothing	4180.23	141	2023-11-29 01:38:12
49	Rest Something	Books	3077.82	181	2023-08-09 03:42:42

5. Display products within a mid-range price bracket (₹2,000 to ₹5,000)
A merchandising team might need this to create a mid-tier pricing campaign.

```
select * from products
where price between 2000 and 5000;
```

product_id	name	category	price	stock_quantity	added_on
2	Population Social	Clothing	4813.68	84	2025-05-30 10:02:50
3	Available Answer	Electronics	2529.51	101	2025-04-13 01:11:46
4	Any Question	Clothing	4759.28	179	2025-06-03 13:34:03
5	Natural Network	Toys	4722.66	75	2023-11-06 00:47:37
7	Response Indeed	Clothing	4897.36	36	2025-03-29 02:43:08
8	Every Amount	Home	4173.60	156	2025-04-30 03:11:10
10	Development System	Electronics	4801.78	153	2025-03-12 08:22:57
12	Action Ask	Electronics	4017.01	19	2025-02-14 03:38:06
13	Full West	Books	2112.33	172	2023-09-15 03:13:38
17	Everything Plant	Books	2496.68	120	2023-10-08 20:11:55
18	Some Them	Toys	3673.86	110	2024-07-25 00:33:53
19	Build High	Clothing	4707.14	47	2023-09-01 02:50:01
20	Real Source	Books	4398.66	197	2025-02-08 10:28:27
21	Bed Including	Clothing	3845.31	92	2025-01-21 11:52:40
22	Move Small	Books	4168.08	22	2023-11-12 18:26:30
23	Life Series	Clothing	2208.99	136	2024-03-09 09:19:13
24	Assume Serve	Home	3437.81	10	2024-07-10 05:05:17
25	South Free	Clothing	3543.58	44	2023-07-25 04:13:43
26	Movement Future	Electronics	3502.62	79	2024-01-23 08:39:07
28	East Foot	Toys	2414.93	128	2025-05-11 16:02:29
29	Drop Remember	Electronics	4160.45	89	2023-12-15 12:31:11
30	Future Sort	Home	3043.67	100	2025-05-17 06:55:28
31	Series Page	Electronics	2070.37	83	2024-04-01 00:24:06
32	Factor Where	Clothing	2295.14	167	2023-07-07 00:59:39
33	Involve Officer	Clothing	4244.09	112	2024-03-22 11:36:17
35	Fire Often	Electronics	4734.89	198	2023-10-22 04:15:51
38	Study Total	Toys	4413.68	31	2023-11-08 08:25:07
39	Data Add	Books	4063.38	62	2025-05-26 23:39:16
41	Serious Recognize	Electronics	4523.10	112	2023-08-20 06:16:55
42	Weight Enter	Clothing	3875.18	60	2024-01-26 08:21:55
45	Suffer We	Electronics	3657.43	113	2024-07-25 12:40:57
46	Age Treatment	Home	3728.83	160	2023-12-01 10:51:52
48	Group But	Clothing	4180.23	141	2023-11-29 01:38:12
49	Rest Something	Books	3077.82	181	2023-08-09 03:42:42
NULL	NULL	NULL	NULL	NULL	NULL

6. Fetch data for specific customer IDs (e.g., from loyalty program list)
This is used when customer IDs are pre-selected from another system.

```
select * from customers
where customer_id in (1,5,10);
```

customer_id	name	email	phone	created_at
1	Thomas Owens	user1@example.com	142-479-1945	2024-10-14 16:01:12
5	David Allen	user5@example.com	(751)456-8289x1	2023-10-29 02:43:00
10	Shawn Hill	user10@example.com	(268)113-3152x7	2023-12-14 20:46:43
NULL	NULL	NULL	NULL	NULL

7. Identify customers whose names start with the letter ‘A’
Used for alphabetical segmentation in outreach or app display

```
select * from customers
where name like "a%";
```

customer_id	name	email	phone	created_at
15	Austin Flores	user15@example.com	329.901.1576x66	2024-06-13 09:03:42
16	Amy Landry	user16@example.com	+1-278-019-3748	2024-02-28 17:51:50
19	Amanda Bright	user19@example.com	380.981.9798x69	2024-12-20 22:58:15
27	Adrienne Green	user27@example.com	530.644.8455x93	2023-08-22 01:55:29
NULL	NULL	NULL	NULL	NULL

8. List electronics products priced under ₹3,000
Used by merchandising or frontend teams to showcase budget electronics.

```
select * from products
where category = "electronics"
and price <3000;
```

product_id	name	category	price	stock_quantity	added_on
3	Available Answer	Electronics	2529.51	101	2025-04-13 01:11:46
6	If Whatever	Electronics	177.40	64	2024-12-19 10:37:14
15	Place Low	Electronics	723.97	46	2023-07-05 14:36:07
31	Series Page	Electronics	2070.37	83	2024-04-01 00:24:06
34	Despite Win	Electronics	1340.34	64	2024-11-27 06:55:45
44	Actually Term	Electronics	396.11	85	2023-11-02 13:09:20
47	Southern Thing	Electronics	512.46	40	2024-02-28 17:57:38
NULL	NULL	NULL	NULL	NULL	NULL

9. Display product names and prices in descending order of price
This helps teams easily view and compare top-priced items.

```
select name, price from products
order by price desc;
```

name	price
Response Indeed	4897.36
Population Social	4813.68
Development System	4801.78
Any Question	4759.28
Fire Often	4734.89
Natural Network	4722.66
Build High	4707.14
Serious Recognize	4523.10
Study Total	4413.68
Real Source	4398.66
Involve Officer	4244.09
Group But	4180.23
Every Amount	4173.60
Move Small	4168.08
Drop Remember	4160.45
Data Add	4063.38
Action Ask	4017.01
Weight Enter	3875.18
Bed Including	3845.31
Age Treatment	3728.83

name	price
Some Them	3673.86
Suffer We	3657.43
South Free	3543.58
Movement Future	3502.62
Assume Serve	3437.81
Rest Something	3077.82
Future Sort	3043.67
Available Answer	2529.51
Everything Plant	2496.68
East Foot	2414.93
Factor Where	2295.14
Life Series	2208.99
Full West	2112.33
Series Page	2070.37
Build Her	1852.64
Between Up	1429.45
Least Green	1398.26
Despite Win	1340.34
Special Fact	1094.18
Answer But	1016.68

name	price
Common Study	985.19
Toward Minute	857.85
Stock Article	834.75
Place Low	723.97
Plant No	639.43
Southern Thing	512.46
Television Stock	421.73
Actually Term	396.11
Listen Development	296.17
If Whatever	177.40

10. Display product names and prices, sorted by price and then by name

The merchandising or catalog team may want to list products from most expensive to cheapest. If multiple products have the same price, they should be sorted alphabetically for clarity on storefronts or printed catalogs.

```
select name, price from products
order by price desc, name asc;
```

name	price
Response Indeed	4897.36
Population Social	4813.68
Development System	4801.78
Any Question	4759.28
Fire Often	4734.89
Natural Network	4722.66
Build High	4707.14
Serious Recognize	4523.10
Study Total	4413.68
Real Source	4398.66

name	price
Involve Officer	4244.09
Group But	4180.23
Every Amount	4173.60
Move Small	4168.08
Drop Remember	4160.45
Data Add	4063.38
Action Ask	4017.01
Weight Enter	3875.18
Bed Including	3845.31
Age Treatment	3728.83
Some Them	3673.86
Suffer We	3657.43
South Free	3543.58
Movement Future	3502.62
Assume Serve	3437.81
Rest Something	3077.82
Future Sort	3043.67
Available Answer	2529.51
Everything Plant	2496.68
East Foot	2414.93

name	price
Factor Where	2295.14
Life Series	2208.99
Full West	2112.33
Series Page	2070.37
Build Her	1852.64
Between Up	1429.45
Least Green	1398.26
Despite Win	1340.34
Special Fact	1094.18
Answer But	1016.68
Common Study	985.19
Toward Minute	857.85
Stock Article	834.75
Place Low	723.97
Plant No	639.43
Southern Thing	512.46
Television Stock	421.73
Actually Term	396.11
Listen Development	296.17
If Whatever	177.40

Level 2: Filtering and Formatting

1. Retrieve orders where customer information is missing (possibly due to data migration or deletion)

Used to identify orphaned orders or test data where customer id is not linked.

```
Select o. * from orders o
Left join customers c
On o .customer id = c. customer id
Where c. customer id is null;
```

order_id	customer_id	order_date	status	total_amount

2. Display customer names and emails using column aliases for frontend readability

Useful for feeding into frontend displays or report headings that require user-friendly labels.

```
Select name as customer name,
Email as customer email
From customers;
```

customer_name	customer_email
Thomas Owens	user1@example.com
Charles Grant	user2@example.com
Kaitlin Richards	user3@example.com
Christina Williams	user4@example.com
David Allen	user5@example.com
Mark Duke	user6@example.com
Briana Wright	user7@example.com
John Bryan	user8@example.com
Jason Thompson	user9@example.com
Shawn Hill	user10@example.com
Walter Jenkins	user11@example.com
Mary Knight	user12@example.com
Leslie Wilson	user13@example.com
Deborah Arias	user14@example.com
Austin Flores	user15@example.com
Amy Landry	user16@example.com
Randy Mooney	user17@example.com
Jeffrey Bray	user18@example.com
Amanda Bright	user19@example.com

customer_name	customer_email
Megan Lee	user20@example.com
Jessica Zamora	user21@example.com
Peter Phillips	user22@example.com
Kathryn Mathews	user23@example.com
Brandy Wright	user24@example.com
Cindy Hart	user25@example.com
Victoria Hall	user26@example.com
Adrienne Green	user27@example.com
Joseph Stuart	user28@example.com
Kara Zavala	user29@example.com
Victoria Acevedo	user30@example.com

3. Calculate total value per item ordered by multiplying quantity and item price

This can help generate per-line item bill details or invoice breakdowns.

```
Select order item id,
Order id,
Product id,
quantity,
item price,
quantity * item price as total value
from order items;
```

order_item_id	order_id	product_id	quantity	item_price	total_value
1	1	19	2	4707.14	9414.28
2	2	6	3	177.40	532.20
3	3	9	3	985.19	2955.57
4	3	23	1	2208.99	2208.99
5	4	35	2	4734.89	9469.78
6	5	19	1	4707.14	4707.14
7	5	7	2	4897.36	9794.72
8	6	7	3	4897.36	14692.08
9	6	37	1	1429.45	1429.45
10	6	15	1	723.97	723.97
11	6	35	3	4734.89	14204.67
12	7	30	1	3043.67	3043.67
13	8	10	3	4801.78	14405.34
14	8	5	3	4722.66	14167.98
15	8	31	2	2070.37	4140.74
16	9	7	3	4897.36	14692.08
17	9	44	1	396.11	396.11
18	9	30	3	3043.67	9131.01
19	10	37	3	1429.45	4288.35

order_item_id	order_id	product_id	quantity	item_price	total_value
20	10	1	1	639.43	639.43
21	10	7	3	4897.36	14692.08
22	10	5	1	4722.66	4722.66
23	11	1	3	639.43	1918.29
24	11	4	3	4759.28	14277.84
25	12	17	2	2496.68	4993.36
26	12	7	1	4897.36	4897.36
27	13	2	2	4813.68	9627.36
28	14	16	3	1094.18	3282.54
29	14	8	3	4173.60	12520.80
30	15	33	2	4244.09	8488.18
31	15	37	1	1429.45	1429.45
32	15	50	3	834.75	2504.25
33	16	38	1	4413.68	4413.68
34	16	50	1	834.75	834.75
35	16	28	2	2414.93	4829.86
36	16	48	2	4180.23	8360.46
37	16	23	2	2208.99	4417.98
38	17	32	3	2295.14	6885.42

order_item_id	order_id	product_id	quantity	item_price	total_value
39	17	37	3	1429.45	4288.35
40	18	19	1	4707.14	4707.14
41	18	14	3	296.17	888.51
42	18	30	3	3043.67	9131.01
43	18	41	3	4523.10	13569.30
44	18	11	2	1852.64	3705.28
45	19	36	2	421.73	843.46
46	20	13	1	2112.33	2112.33
47	20	37	3	1429.45	4288.35
48	21	19	2	4707.14	9414.28
49	21	30	1	3043.67	3043.67
50	21	33	2	4244.09	8488.18
51	21	23	2	2208.99	4417.98
52	22	11	2	1852.64	3705.28
53	22	42	2	3875.18	7750.36
54	22	14	2	296.17	592.34
55	22	49	3	3077.82	9233.46
56	23	33	1	4244.09	4244.09
57	23	2	2	4813.68	9627.36

order_item_id	order_id	product_id	quantity	item_price	total_value
58	23	8	1	4173.60	4173.60
59	24	34	2	1340.34	2680.68
60	24	6	1	177.40	177.40
61	24	21	3	3845.31	11535.93
62	24	33	2	4244.09	8488.18
63	25	47	3	512.46	1537.38
64	25	46	2	3728.83	7457.66
65	25	9	1	985.19	985.19
66	25	36	3	421.73	1265.19
67	25	22	3	4168.08	12504.24
68	26	42	3	3875.18	11625.54
69	26	18	3	3673.86	11021.58
70	26	8	3	4173.60	12520.80
71	27	9	3	985.19	2955.57
72	28	1	1	639.43	639.43
73	29	45	2	3657.43	7314.86
74	29	8	2	4173.60	8347.20
75	29	21	2	3845.31	7690.62
76	29	9	3	985.19	2955.57

order_item_id	order_id	product_id	quantity	item_price	total_value
77	30	23	3	2208.99	6626.97
78	30	9	3	985.19	2955.57
79	30	41	2	4523.10	9046.20
80	31	43	3	1398.26	4194.78
81	31	47	2	512.46	1024.92
82	31	21	2	3845.31	7690.62
83	31	17	1	2496.68	2496.68
84	31	28	3	2414.93	7244.79
85	32	30	3	3043.67	9131.01
86	32	10	3	4801.78	14405.34
87	33	20	1	4398.66	4398.66
88	33	19	2	4707.14	9414.28
89	33	23	3	2208.99	6626.97
90	33	45	3	3657.43	10972.29
91	33	35	1	4734.89	4734.89
92	34	8	1	4173.60	4173.60
93	34	50	2	834.75	1669.50
94	34	17	3	2496.68	7490.04
95	34	44	3	396.11	1188.33
96	34	23	3	2208.99	6626.97
97	35	35	2	4734.89	9469.78
98	36	12	2	4017.01	8034.02
99	36	14	1	296.17	296.17
100	36	29	1	4160.45	4160.45
101	36	30	3	3043.67	9131.01
102	36	22	3	4168.08	12504.24
103	37	1	1	639.43	639.43
104	38	24	2	3437.81	6875.62
105	38	41	1	4523.10	4523.10
106	38	16	3	1094.18	3282.54
107	39	33	1	4244.09	4244.09
108	39	37	2	1429.45	2858.90
109	39	22	2	4168.08	8336.16
110	40	24	3	3437.81	10313.43
111	40	30	2	3043.67	6087.34
112	40	9	1	985.19	985.19
113	40	4	1	4759.28	4759.28
114	40	50	3	834.75	2504.25

order_item_id	order_id	product_id	quantity	item_price	total_value
115	41	15	1	723.97	723.97
116	42	21	3	3845.31	11535.93
117	42	42	1	3875.18	3875.18
118	42	31	3	2070.37	6211.11
119	43	11	2	1852.64	3705.28
120	43	37	2	1429.45	2858.90
121	43	29	3	4160.45	12481.35
122	44	47	2	512.46	1024.92
123	44	29	3	4160.45	12481.35
124	44	48	1	4180.23	4180.23
125	45	27	3	1016.68	3050.04
126	45	2	2	4813.68	9627.36
127	45	18	2	3673.86	7347.72
128	45	23	2	2208.99	4417.98
129	45	40	3	857.85	2573.55
130	46	10	2	4801.78	9603.56
131	46	23	2	2208.99	4417.98
132	46	41	2	4523.10	9046.20
133	46	34	1	1340.34	1340.34

order_item_id	order_id	product_id	quantity	item_price	total_value
134	47	27	1	1016.68	1016.68
135	48	41	3	4523.10	13569.30
136	48	15	2	723.97	1447.94
137	48	34	2	1340.34	2680.68
138	48	18	3	3673.86	11021.58
139	49	46	1	3728.83	3728.83
140	49	1	2	639.43	1278.86
141	49	49	2	3077.82	6155.64
142	49	39	3	4063.38	12190.14
143	50	35	3	4734.89	14204.67
144	51	4	3	4759.28	14277.84
145	51	30	1	3043.67	3043.67
146	51	17	3	2496.68	7490.04
147	52	34	1	1340.34	1340.34
148	52	20	2	4398.66	8797.32
149	52	32	3	2295.14	6885.42
150	52	29	3	4160.45	12481.35
151	52	47	2	512.46	1024.92
152	53	32	1	2295.14	2295.14

order_item_id	order_id	product_id	quantity	item_price	total_value
153	53	28	2	2414.93	4829.86
154	54	3	1	2529.51	2529.51
155	54	15	3	723.97	2171.91
156	54	17	1	2496.68	2496.68
157	55	10	3	4801.78	14405.34
158	55	23	3	2208.99	6626.97
159	55	47	3	512.46	1537.38
160	55	27	2	1016.68	2033.36
161	55	43	1	1398.26	1398.26
162	56	38	1	4413.68	4413.68
163	56	28	1	2414.93	2414.93
164	57	34	2	1340.34	2680.68
165	58	13	2	2112.33	4224.66
166	58	11	3	1852.64	5557.92
167	59	40	1	857.85	857.85
168	60	49	1	3077.82	3077.82
169	60	6	1	177.40	177.40
170	60	9	2	985.19	1970.38
171	60	34	3	1340.34	4021.02

order_item_id	order_id	product_id	quantity	item_price	total_value
172	61	8	3	4173.60	12520.80
173	61	6	3	177.40	532.20
174	62	42	3	3875.18	11625.54
175	62	43	3	1398.26	4194.78
176	62	27	3	1016.68	3050.04
177	62	10	1	4801.78	4801.78
178	62	12	3	4017.01	12051.03
179	63	2	3	4813.68	14441.04
180	63	19	1	4707.14	4707.14
181	63	8	3	4173.60	12520.80
182	63	31	1	2070.37	2070.37
183	63	17	1	2496.68	2496.68
184	64	16	2	1094.18	2188.36
185	64	23	2	2208.99	4417.98
186	64	33	3	4244.09	12732.27
187	64	27	3	1016.68	3050.04
188	64	50	3	834.75	2504.25
189	65	32	1	2295.14	2295.14
190	65	50	2	834.75	1669.50

order_item_id	order_id	product_id	quantity	item_price	total_value
191	66	7	1	4897.36	4897.36
192	66	8	3	4173.60	12520.80
193	66	36	1	421.73	421.73
194	66	22	2	4168.08	8336.16
195	67	40	2	857.85	1715.70
196	67	3	1	2529.51	2529.51
197	67	20	3	4398.66	13195.98
198	67	11	2	1852.64	3705.28
199	68	26	1	3502.62	3502.62
200	68	10	3	4801.78	14405.34
201	68	48	2	4180.23	8360.46
202	68	16	1	1094.18	1094.18
203	68	20	2	4398.66	8797.32
204	69	38	3	4413.68	13241.04
205	69	10	1	4801.78	4801.78
206	70	38	2	4413.68	8827.36
207	71	16	3	1094.18	3282.54
208	71	48	3	4180.23	12540.69
209	71	15	2	723.97	1447.94

order_item_id	order_id	product_id	quantity	item_price	total_value
210	71	1	3	639.43	1918.29
211	72	29	1	4160.45	4160.45
212	73	24	1	3437.81	3437.81
213	73	41	1	4523.10	4523.10
214	73	22	3	4168.08	12504.24
215	73	6	2	177.40	354.80
216	74	15	1	723.97	723.97
217	74	24	2	3437.81	6875.62
218	74	46	2	3728.83	7457.66
219	74	16	2	1094.18	2188.36
220	74	50	2	834.75	1669.50
221	75	10	1	4801.78	4801.78
222	75	7	2	4897.36	9794.72
223	75	1	2	639.43	1278.86
224	76	30	1	3043.67	3043.67
225	76	6	2	177.40	354.80
226	76	25	2	3543.58	7087.16
227	76	4	2	4759.28	9518.56
228	76	26	1	3502.62	3502.62

Remark:- has been 1200 row not possible all screenshot so, I have a some screenshot.

4. Combine customer name and phone number in a single column

Used to show brief customer summaries or contact lists.

```
Select concat (name,"-", phone) as
Customers contact from customers;
```

customers_contact
Thomas Owens-142-479-1945
Charles Grant-9153947511
Kaitlin Richards-2073473421
Christina Williams-586-605-5061x06
David Allen-(751)456-8289x1
Mark Duke-(144)957-2811
Briana Wright-223-833-9635
John Bryan-045.568.0798x27
Jason Thompson-1862659420
Shawn Hill-(268)113-3152x7
Walter Jenkins-536-329-0817x71
Mary Knight-361-636-3802
Leslie Wilson-+1-256-261-1984
Deborah Arias-811-821-2144x97
Austin Flores-329.901.1576x66
Amy Landry-+1-278-019-3748
Randy Mooney-(158)927-7313x3
Jeffrey Bray-164.962.0222x92
Amanda Bright-380.981.9798x69

customers_contact
Megan Lee-7044478333
Jessica Zamora-177-816-7773x00
Peter Phillips-389-656-1695
Kathryn Mathews-001-707-722-178
Brandy Wright-853.520.8915x61
Cindy Hart-617.574.8421x41
Victoria Hall-926-840-0016
Adrienne Green-530.644.8455x93
Joseph Stuart-363-045-4287
Kara Zavala-434-985-5776x61
Victoria Acevedo-+1-325-063-7244

5. Extract only the date part from order timestamps for date-wise reporting

Helps group or filter orders by date without considering time.

```
Select order id,
Date (order date) as
Order date only
From orders;
```

order_id	order_date_only
1	2025-03-02
2	2024-10-09
3	2025-05-08
4	2024-09-19
5	2025-04-08
6	2024-10-25
7	2024-07-29
8	2024-07-30
9	2025-06-10
10	2025-02-16
11	2025-03-11
12	2025-01-15
13	2025-05-12
14	2024-09-24
15	2024-07-18
16	2024-09-09
17	2024-08-19
18	2024-06-15

6. List products that do not have any stock left

This helps the inventory team identify out-of-stock items.

```
Select * from products
Where stock quantity=0;
```

product_id	name	category	price	stock_quantity	added_on
NULL	NULL	NULL	NULL	NULL	NULL

Level 3: Aggregations

1. Count the total number of orders placed

Used by business managers to track order volume over time.

```
Select count (*) as total orders
From orders;
```

total_orders
400

2. Calculate the total revenue collected from all orders

This gives the overall sales value.

```
Select sum (total amount) as
Total revenue
From orders;
```

total_revenue
6960973.66

3. Calculate the average order value

Used for understanding customer spending patterns.

```
Select avg (total amount) as
Average order value
From orders;
```

average_order_value
17402.434150

4. Count the number of customers who have placed at least one order

This identifies active customers.

```
Select count (distinct customer id)as
Customers with orders
From orders;
```

customers_with_orders
30

5. Find the number of orders placed by each customer

Helpful for identifying top or repeat customers.

```
Select customer id, count (order id)
As total orders
From orders
Group by customer id;
```

customer_id	total_orders
1	12
2	17
3	17
4	9
5	14
6	16
7	13
8	10
9	10
10	13
11	9
12	11
13	12
14	15
15	15

6. Find total sales amount made by each customer

```
Select customer id,
Sum (total amount) as total sales
From orders
Group by customer id;
```

customer_id	total_sales
1	183747.44
2	284420.07
3	253783.31
4	137562.22
5	262504.19
6	212173.58
7	167960.11
8	164701.78
9	106226.03
10	252722.75
11	173409.01
12	169653.79
13	152727.70
14	360324.18

7. List the number of products sold per category

This helps category managers assess performance by department.

```
Select p. category, sum (oi. quantity)
As total products sold
From products p
Join order items oi
On p. product id = oi .product id
Group by p. category;
```

category	total_products_sold
Home	443
Clothing	559
Electronics	687
Toys	405
Books	350

8. Find the average item price per category

Useful to compare pricing across departments.

```
Select category, avg (price) as average item price
From products
Group by category;
```

category	average_item_price
Home	2146.367500
Clothing	3434.581538
Electronics	2653.388571
Toys	2516.526250
Books	3167.084286

9. Show number of orders placed per day

Used to track daily business activity and demand trends.

```
Select date (order date) as order day,
Count (order id) as total orders
From orders
Group by date (order date)
Order by order day;
```

order_day	total_orders
2024-06-15	1
2024-06-18	1
2024-06-19	3
2024-06-20	2
2024-06-21	2
2024-06-23	2
2024-06-24	1
2024-06-25	1
2024-06-30	2
2024-07-03	1
2024-07-04	1
2024-07-05	3
2024-07-06	2
2024-07-08	2

10. List total payments received per payment method

Helps the finance team understand preferred transaction modes.

```
Select method,
Sum (amount paid)
As total payment received
From payments
Group by method;
```

method	total_payment_received
Credit Card	1754603.10
Net Banking	1658383.90
UPI	1617408.78
Debit Card	1930577.88

Level 4: Multi-Table Queries (JOINS)

1. Retrieve order details along with the customer name (INNER JOIN)

Used for displaying which customer placed each order.

```
Select o. order id,
o. order date,
o. total amount,
c. name as customer name
from orders o
inner join customers c
on o. customer id = c. customer id;
```

order_id	order_date	total_amount	customer_name
14	2024-09-24 21:21:38	15803.34	Thomas Owens
17	2024-08-19 21:17:57	11173.77	Thomas Owens
61	2024-12-26 23:21:58	13053.00	Thomas Owens
76	2025-01-14 22:59:54	23506.81	Thomas Owens
92	2024-09-26 11:33:58	834.75	Thomas Owens
109	2025-03-17 04:56:18	12190.14	Thomas Owens
127	2025-06-10 18:01:32	20160.64	Thomas Owens
135	2025-03-11 22:58:23	8891.79	Thomas Owens
144	2024-09-19 09:18:22	12093.54	Thomas Owens
221	2024-09-25 05:49:39	12437.61	Thomas Owens
278	2025-02-01 20:05:38	32015.16	Thomas Owens
379	2025-03-22 15:57:58	21586.89	Thomas Owens
56	2024-06-23 15:59:57	6828.61	Charles Grant

2. Get list of products that have been sold (INNER JOIN with order items)

Used to find which products were actually included in orders.

```
Select distinct p. product id,
p.name,
p. category,
p. price
from products p
inner join order items oi
on p. product id = oi. Product id;
```

product_id	name	category	price
1	Plant No	Home	639.43
2	Population Social	Clothing	4813.68
3	Available Answer	Electronics	2529.51
4	Any Question	Clothing	4759.28
5	Natural Network	Toys	4722.66
6	If Whatever	Electronics	177.40
7	Response Indeed	Clothing	4897.36
8	Every Amount	Home	4173.60
9	Common Study	Toys	985.19
10	Development System	Electronics	4801.78
11	Build Her	Books	1852.64
12	Action Ask	Electronics	4017.01
13	Full West	Books	2112.33

3. List all orders with their payment method (INNER JOIN)

Used by finance or audit teams to see how each order was paid for.

```
Select o. order id,
o. order date,
p. method
as payment method
from orders o
inner join payments p
on o. order id = p. order id;
```

order_id	order_date	payment_method
1	2025-03-02 07:20:11	Credit Card
2	2024-10-09 18:08:21	Net Banking
3	2025-05-08 00:08:27	Credit Card
4	2024-09-19 22:16:13	UPI
5	2025-04-08 18:02:06	UPI
6	2024-10-25 07:33:59	UPI
7	2024-07-29 11:58:47	UPI
8	2024-07-30 22:49:49	Net Banking
9	2025-06-10 17:00:25	Net Banking
10	2025-02-16 12:45:59	Debit Card
11	2025-03-11 14:22:46	Credit Card
12	2025-01-15 13:22:53	Credit Card
13	2025-05-12 13:04:29	UPI

4. Get list of customers and their orders (LEFT JOIN)

Used to find all customers and see who has or hasn’t placed orders.

```
select c. customer id,
c.name as customer name,
o. order id,
o. order date,
o. total amount
from customers c
left join orders o
on c. customer id = o. customer id;
```

customer_id	customer_name	order_id	order_date	total_amount
1	Thomas Owens	14	2024-09-24 21:21:38	15803.34
1	Thomas Owens	17	2024-08-19 21:17:57	11173.77
1	Thomas Owens	61	2024-12-26 23:21:58	13053.00
1	Thomas Owens	76	2025-01-14 22:59:54	23506.81
1	Thomas Owens	92	2024-09-26 11:33:58	834.75
1	Thomas Owens	109	2025-03-17 04:56:18	12190.14
1	Thomas Owens	127	2025-06-10 18:01:32	20160.64
1	Thomas Owens	135	2025-03-11 22:58:23	8891.79
1	Thomas Owens	144	2024-09-19 09:18:22	12093.54
1	Thomas Owens	221	2024-09-25 05:49:39	12437.61
1	Thomas Owens	278	2025-02-01 20:05:38	32015.16
1	Thomas Owens	379	2025-03-22 15:57:58	21586.89
2	Charles Grant	56	2024-06-23 15:59:57	6828.61

5. List all products along with order item quantity (LEFT JOIN)

Useful for inventory teams to track what sold and what hasn't .

```
Select p. product id,
p.name
as product name,
oi. quantity
from products p
left join order items oi
on p. product id = oi. product id;
```

product_id	product_name	quantity
1	Plant No	1
1	Plant No	3
1	Plant No	1
1	Plant No	1
1	Plant No	2
1	Plant No	3
1	Plant No	2
1	Plant No	2
1	Plant No	2
1	Plant No	2
1	Plant No	2
1	Plant No	3
1	Plant No	3

6. List all payments including those with no matching orders (RIGHT JOIN)

Rare but used when ensuring all payments are mapped correctly.

```
Select p. payment  id,
p. order id,
p. method,
p. amount paid, o. order id
as matched order id
from orders as o
right join payments as p
on o. order id = p. order id;
```

payment_id	order_id	method	amount_paid	matched_order_id
1	1	Credit Card	9414.28	1
2	2	Net Banking	532.20	2
3	3	Credit Card	5164.56	3
4	4	UPI	9469.78	4
5	5	UPI	14501.86	5
6	6	UPI	31050.17	6
7	7	UPI	3043.67	7
8	8	Net Banking	32714.06	8

7. Combine data from three tables: customer, order, and payment

Used for detailed transaction reports.

```
Select c. customer id,
c.name, o. order id,
o. order date, p. payment id,
p. method, p. amount paid
from customers as c
inner join orders as o
on c. customer id = o. customer id
inner join payments as p
on o. order id = p. order id;
```

payment_id	order_id	method	amount_paid	matched_order_id
1	1	Credit Card	9414.28	1
2	2	Net Banking	532.20	2
3	3	Credit Card	5164.56	3
4	4	UPI	9469.78	4
5	5	UPI	14501.86	5
6	6	UPI	31050.17	6
7	7	UPI	3043.67	7
8	8	Net Banking	32714.06	8
9	9	Net Banking	24219.20	9
10	10	Debit Card	24342.52	10
11	11	Credit Card	16196.13	11
12	12	Credit Card	9890.72	12
13	13	UPI	9627.36	13
14	14	UPI	15803.34	14

Level 5: Subqueries (Inner Queries)

1. List all products priced above the average product price

Used by pricing analysts to identify premium-priced products.

```
Select * from products
Where price > (select avg (price)
From products);
```

product_id	name	category	price	stock_quantity	added_on
2	Population Social	Clothing	4813.68	84	2025-05-30 10:02:50
4	Any Question	Clothing	4759.28	179	2025-06-03 13:34:03
5	Natural Network	Toys	4722.66	75	2023-11-06 00:47:37
7	Response Indeed	Clothing	4897.36	36	2025-03-29 02:43:08
8	Every Amount	Home	4173.60	156	2025-04-30 03:11:10
10	Development System	Electronics	4801.78	153	2025-03-12 08:22:57
12	Action Ask	Electronics	4017.01	19	2025-02-14 03:38:06
18	Some Them	Toys	3673.86	110	2024-07-25 00:33:53
19	Build High	Clothing	4707.14	47	2023-09-01 02:50:01
20	Real Source	Books	4398.66	197	2025-02-08 10:28:27

2. Find customers who have placed at least one order

Used to identify active customers for loyalty campaigns.

```
Select * from customers where customer id
In (select customer id from orders);
```

customer_id	name	email	phone	created_at
1	Thomas Owens	user1@example.com	142-479-1945	2024-10-14 16:01:12
2	Charles Grant	user2@example.com	9153947511	2023-11-25 15:45:24
3	Kaitlin Richards	user3@example.com	2073473421	2024-06-23 09:55:22
4	Christina Williams	user4@example.com	586-605-5061x06	2024-10-27 17:19:38
5	David Allen	user5@example.com	(751)456-8289x1	2023-10-29 02:43:00
6	Mark Duke	user6@example.com	(144)957-2811	2024-06-24 03:22:59
7	Briana Wright	user7@example.com	223-833-9635	2023-06-25 00:35:43
8	John Bryan	user8@example.com	045.568.0798x27	2025-02-15 17:57:04
9	Jason Thompson	user9@example.com	1862659420	2024-08-31 08:18:51
10	Shawn Hill	user10@example.com	(268)113-3152x7	2023-12-14 20:46:43

3. Show orders whose total amount is above the average for that customer

Used to detect unusually high purchases per customer.

```
Select * from orders o
Where total amount > (select avg (total amount) from orders
Where customer id = o. customer id);
```

order_id	customer_id	order_date	status	total_amount
6	29	2024-10-25 07:33:59	Cancelled	31050.17
8	19	2024-07-30 22:49:49	Cancelled	32714.06
9	6	2025-06-10 17:00:25	Pending	24219.20
10	28	2025-02-16 12:45:59	Delivered	24342.52
14	1	2024-09-24 21:21:38	Shipped	15803.34
16	5	2024-09-09 03:00:40	Shipped	22856.73
18	13	2024-06-15 12:57:04	Cancelled	32001.24
21	22	2025-03-12 18:53:46	Pending	25364.11
22	20	2025-02-21 11:21:06	Shipped	21281.44
24	6	2024-06-19 07:09:04	Pending	22882.19

4. Display customers who haven't placed any orders

Used for re-engagement campaigns targeting inactive users.

```
Select * from customers where customer id
Not in (select customer id from orders);
```

customer_id	name	email	phone	created_at
NULL	NULL	NULL	NULL	NULL

5. Show products that were never ordered

Helps with inventory clearance decisions or product deactivation.

```
Select * from products where product id
Not in (select product id from order items);
```

product_id	name	category	price	stock_quantity	added_on
NULL	NULL	NULL	NULL	NULL	NULL

6. Show highest value order per customer

Used to identify the largest transaction made by each customer.

```
Select * from orders o where total amount = (select max (total amount)
From orders where customer id = o. customer id);
```

order_id	customer_id	order_date	status	total_amount
18	13	2024-06-15 12:57:04	Cancelled	32001.24
33	8	2025-04-27 01:26:11	Pending	36147.09
62	10	2025-06-13 23:21:01	Cancelled	35723.17
66	27	2024-06-19 09:41:16	Pending	26176.05
68	24	2025-03-15 07:27:20	Delivered	36159.92
80	12	2025-03-28 18:42:12	Delivered	38656.26
107	14	2025-02-01 02:02:18	Shipped	48042.06
121	16	2024-11-21 16:50:01	Pending	37415.67
128	7	2025-05-05 16:49:18	Pending	28589.04
133	11	2024-11-20 00:44:46	Shipped	37129.82

7. Highest Order Per Customer (Including Names)

Used to identify the largest transaction made by each customer. Outputs name as well.

```
Select c.name, o.* from customers c inner join orders o

On c. customer id = o. customer id

Where o. total amount = (select max (o2.total_amount)

From orders o2

Where o2.customer_id= o. customer id);
```

name	order_id	customer_id	order_date	status	total_amount
Thomas Owens	278	1	2025-02-01 20:05:38	Delivered	32015.16
Charles Grant	315	2	2025-01-27 15:23:27	Cancelled	42056.04
Kaitlin Richards	281	3	2024-11-08 07:40:55	Shipped	41679.11
Christina Williams	386	4	2024-12-18 11:35:41	Cancelled	25747.34
David Allen	266	5	2025-03-21 21:25:00	Delivered	39921.78
Mark Duke	330	6	2025-01-10 15:50:45	Pending	39003.19
Briana Wright	128	7	2025-05-05 16:49:18	Pending	28589.04
John Bryan	33	8	2025-04-27 01:26:11	Pending	36147.09
Jason Thompson	348	9	2024-12-29 18:35:10	Cancelled	21414.44
Shawn Hill	62	10	2025-06-13 23:21:01	Cancelled	35723.17

Level 6: Set Operations

1. List all customers who have either placed an order or written a product review

Used to identify engaged customers from different activity areas.

```
select customer_id, name

from customers where customer_id in (select customer_id from orders union select customer_id
from product_reviews);
```

customer_id	name
1	Thomas Owens
2	Charles Grant
3	Kaitlin Richards
4	Christina Williams
5	David Allen
6	Mark Duke
7	Briana Wright
8	John Bryan
9	Jason Thompson
10	Shawn Hill

2. List all customers who have placed an order as well as reviewed a product [intersect not supported]

Used to identify highly engaged customers for rewards.

```
select c.customer_id, c.name from customers c where exists (select 1 from orders o where o.
customer_id= c.customer_id)

and exists (select 1 from product_reviews r where r.customer_id = c.customer_id);
```

customer_id	name
1	Thomas Owens
2	Charles Grant
4	Christina Williams
6	Mark Duke
7	Briana Wright
9	Jason Thompson
10	Shawn Hill
11	Walter Jenkins
13	Leslie Wilson
14	Deborah Arias